Part 1: General View

1. Analysing a command line: command, options, arguments.
2. Understanding error messages and getting help.
3. Practising the basic commands: `date`, `cal`, `echo`, `bc`, `history`, `clear`.
4. First steps in the file tree: `pwd`, `ls`, `cd`.

COMMAND [OPTIONS] [ARGUMENTS]

Lab rules: type every command yourself and press Enter; write what you *observe*, then the *role* of the command.
Unix distinguishes upper-case and lower-case letters. Do not use `sudo`.

Part 2: Syntax of a command

Exercise 1: For each command line, give the command, the option(s) and the argument(s), then execute it.

Command line	Command	Option(s)	Argument(s)	Role
`date`				
`date -u`				
`cal 10 2026`				
`cal -3`				
`cal -j 2026`				
`ls -l /home`				
`ls --all /home /tmp`				

Exercise 2: Options.

1. Execute `ls -a -l`, `ls -al` and `ls -la`. What do you notice? .....
2. Execute `ls -a` and `ls --all`. What do you notice? .....
3. Execute `date -u` and `date --utc`: .....
4. What separates the command, the options and the arguments? .....

Exercise 3: Each of the following lines fails. Execute it, copy the message, explain the error and give the correct line.

Line	Message	Explanation	Correct line
`CAL 2026`			
`cal 2026 10`			
`ls-a`			
`date -q`			
`ls /Home`			

1. Execute `date` then `echo $?`: .....
2. Execute `ls /Home` then `echo $?`: .....
3. Execute `date ; whoami ; hostname`. What is the role of `;`? .....
4. Execute `date # today`. What is the role of `#`? .....

Part 3: Getting help

Exercise 4: Execute the following commands, observe the result, and state their roles:

1. `cal --help`: .....
2. `man cal` (quit with `q`): .....
3. `whatis cal`: .....
4. `type date`: .....
5. `type cd`: .....
6. `help cd`: .....
7. With `man cal`, copy the synopsis of the command: .....
8. In a synopsis, what do the square brackets `[ ]` mean? .....
9. With `ls --help`, find the long form of the option `-r`: .....
10. What is the difference between `--help`, `man`, `whatis` and `help`?

    .....

    .....

Part 4: Basic commands

Exercise 5: `date` and `cal`. Execute the following commands, observe the result, and state their roles:

1. `cal`: .....
2. `cal 2026` and `cal -y`: .....
3. `cal 26` and `cal 2026`: what is the difference? .....
4. On which day of the week were you born? Command used: .....
5. `date +%d/%m/%Y`: .....
6. `date +%H:%M`: .....
7. `date +%A`: ..... `date +%j`: .....
8. The command that displays the date as `2026-10-12` is: .....

Exercise 6: `echo`. Execute the following commands, observe the result, and state their roles:

1. `echo University Ferhat Abbas - Setif 1`: .....
2. `echo one two three`: .....
3. `echo "one two three"`: .....
4. `echo "I am $USER"`: .....
5. `echo 'I am $USER'`: .....
6. `echo -n Hello`: .....
7. `echo -e "Name\tGroup"`: .....
8. With a single `echo`, display your last name and your first name on two lines: .....

Exercise 7: `bc`, `history` and `clear`.

1. `bc -q`: .....
   - `3+26`: ..... `68-1235`: ..... `7*8`: ..... `2^10`: .....
   - `17/5`: ..... `17%5`: .....
   - `scale=3` then `17/5`: .....
   - `quit` (or Ctrl+D): .....
2. Why does the prompt `$` disappear inside `bc`? .....
3. `history`: .....
4. `history 5`: .....
5. Press the key ↑ several times. What happens? .....
6. Re-run one command of the list with `!`*number*. Command typed: .....
7. `clear` and `Ctrl+L`: .....

Part 5: First steps in the file tree

Exercise 8: `pwd`, `ls` and `cd`.

1. `pwd`: .....
2. `ls`: .....
3. `ls -a`: .....
4. `ls -l`: .....
5. `ls -lh`: .....
6. `ls -lt` and `ls -ltr`: .....
7. `ls /`: .....
8. `cd /etc` then `pwd`: .....
9. How did the prompt change? .....
10. `cd ..` then `pwd`: .....
11. `cd` then `pwd`: .....
12. `cd -`: .....
13. `cd ~`: .....
14. Go to `/usr`, list its contents in long format, and come back home. Commands typed:
    .....
15. What does `cd` display when it succeeds? .....